

Sample Question Bank - Software System Design

Module 1:

1. Why can't we just use Vertical Scaling indefinitely if hardware keeps getting cheaper according to Moore's Law?
2. Is 100% Availability physically possible in distributed systems?
3. What is the difference between Latency and Response Time?
4. If Horizontal Scaling is so superior, why do companies still build Monoliths?
5. How does caching affect Little's Law ($L = \lambda \cdot W$)?
6. A system experiences 45 minutes of unplanned downtime in a single calendar month. Calculate its operational availability percentage for that month. Did it meet a 99.9% SLA target?
7. If a microservice needs data from another microservice's database, can we just use a SQL JOIN across databases?
8. What is a 'Distributed Monolith' and how do teams accidentally build one?
9. How do you handle user authentication across multiple microservices?
10. How does the Saga pattern solve distributed transactions in microservices?
11. Design a simplified e-commerce application (Users, Catalog, Cart, Orders, Payments, Shipping). Group these features into a Modular Monolith package structure vs. a Microservices Bounded Context layout, detailing which database engine you would select for each microservice domain.
12. Why do we multiply by 8 when calculating network bandwidth, but not when calculating disk storage?
13. What is the difference between an API Gateway and a Load Balancer in an N-Tier architecture?
14. How do you handle a system where Peak QPS is 100 times higher than Average QPS (e.g., ticket sales for a major concert)?
15. Should static images and video files be stored directly inside a relational database like PostgreSQL?
16. Consider an E-Commerce Platform, Calculate the system requirements for an online store with 20 Million Daily Active Users (DAU). a. Calculate Average Read QPS and Peak Read QPS (3 times peak factor). b. Calculate Average Write QPS and Peak Write QPS c. If each order transaction record requires 4 KB of database storage, calculate the required database storage growth per year.
17. A video platform serves 100 Million video metadata views per day. Each metadata payload response is 5 KB. Apply the 80/20 Pareto rule to calculate how many

Gigabytes of RAM are required for the Redis cache cluster to achieve an 80% cache hit ratio.

18. Why does an unindexed database query cause high tail latency?
19. What is the difference between processing latency and network jitter?
20. How does a high cache hit ratio help reduce p99 tail latency for queries that MISS the cache?
21. A real-time bidding API fans out in parallel to $N = 200$ ad-network provider microservices. Each ad-network service has a p99 response SLA of 50 ms 1% probability of taking longer than 50 ms. Calculate the exact probability that an end-user search request will experience tail latency > 50 ms waiting for all 200 ad networks to respond.
22. Design a user profile API endpoint `/api/v1/user/dashboard` with a target p99 latency budget of < 50 ms. **Operations List:**
 - OAuth Token Decryption (CPU): 2 ms
 - Fetch User Core Profile (Redis Read): 1 ms
 - Fetch User Order History (PostgreSQL Read): 35 ms
 - Fetch Recommended Items (ML Engine over gRPC): 40 ms
 - Write User Activity Event (Kafka Producer): 3 ms. Determine if a sequential execution flow satisfies the 50 ms budget. If not, draw an optimized parallelized execution topology that meets the requirement.
23. Explain vertical and horizontal scaling. State its advantages and disadvantages.
24. Differentiate between scale-up and scale-out with one real-world example for each.
25. What is meant by high availability? Define a single point of failure (SPOF)?
26. Differentiate between network latency and processing latency.
27. What is the difference between latency and throughput? Why is low latency important for an interactive application?
28. Compare vertical scaling and horizontal scaling with respect to: Cost, Hardware limitations, Fault tolerance and complexity of the model.
29. Explain how a web application can be horizontally scaled using multiple application servers. Why is horizontal scaling generally important for large-scale distributed systems?
30. Design the high-level architecture of a high-traffic web application and explain how your design addresses scalability and availability.
31. Differentiate between monolithic and microservices architecture.
32. What is back-of-the-envelope estimation? Explain Fermi estimation
33. Explain how QPS can be estimated from the number of daily users and requests per user.
34. A photo-sharing application has **10 million users**, out of which **1 million are daily active users**. Each active user:
 - Views 100 images/day

- Uploads 2 images/day
- Average image size = 1 MB

Calculate:

- a. Number of image-view requests per day
- b. Average image-read QPS
- c. Number of images uploaded per day
- d. Daily storage growth
- e. Annual storage growth
- f. Explain how caching/CDN can affect bandwidth and latency.

35. A social media application has **5 million daily active users**. Each user uploads an average of **2 images per day**, with each image having an average size of **2 MB**. Calculate:

- a. Images generated per day
- b. Storage required per day
- c. Storage required for 30 days
- d. Storage required for one year
- e. Discuss factors such as replication, metadata, and backups that may increase the actual storage requirement.

Module 2:

1. If Layer 4 load balancers are faster and use less CPU, why don't we use them everywhere instead of Layer 7?
2. What happens to active WebSocket connections during load balancer connection draining?
3. "Isn't reading user sessions from an external Redis cluster slower than reading local server RAM? Justify your answer.
4. If a user revokes their account or changes their password, how do you invalidate a stateless JWT token before it expires?
5. Compare between L4 vs. L7 CPU Capacity Sizing
6. A high-traffic platform processes 50,000 incoming HTTPS requests per second during peak hours. **Scenario A:** L4 load balancing requires 0.01 CPU cores per 1,000 QPS. **Scenario B:** L7 load balancing (with TLS decryption and HTTP header parsing) requires 0.25 CPU cores per 1,000 QPS. Calculate the total number of CPU cores required to run the load balancer tier under Scenario A versus Scenario B.

Determine the operational trade-offs of running L4 TLS pass-through versus L7 TLS termination

7. Describe Stateless and Stateful service architectures
8. An enterprise service handles 20,000 QPS on its API tier. Compare the database/cache network bandwidth required for a. Session ID approach: Every request performs a 1 KB session lookup call to a Redis cluster. b. Stateless JWT approach: Every request carries a 1 KB cryptographically signed token validated locally by the app server CPU. Calculate total network ingress/egress bytes per second imposed on the session tier under both approaches.
9. What is the difference between Weighted Round Robin (WRR) and Weighted Least Connections (WLC)?
10. A load balancer uses NGINX Smooth Weighted Round Robin across three servers with weights $W_A = 5$, $W_B = 2$, and $W_C = 1$ (Total Weight = 8). Construct a trace table for the first 8 incoming requests showing:
 - a. Current weight values (CW) before selection.
 - b. Selected server instance for each request.
 - c. Adjusted weight values (CW) after applying the mathematical reduction formula.
11. What is the key functional difference between a Reverse Proxy and an API Gateway?
12. How does HAProxy achieve such high performance if it runs on a single event loop process?
13. Why would an organization deploy an AWS NLB in front of an AWS ALB?
14. How does SSL/TLS Interception in a Forward Proxy avoid triggering browser certificate warnings?
15. Why would I choose Cache-Aside over Read-Through if Read-Through produces cleaner application code?
16. What is the difference between Cache Eviction and Cache Invalidation?
17. How does a Bloom Filter prevent Cache Penetration with zero false negatives?
18. Why does the XFetch algorithm prevent Cache Stampedes better than a simple Mutex Lock?
19. In Write-Back caching, how do system architects prevent data loss if a Redis node crashes before flushing dirty writes to the database?
20. Implement a functional LRU Cache class in Python or C++ with `get(key)` and `put(key, value)` operations. **Task Requirements:**
 - a. Both `get` and `put` must execute in $O(1)$ time complexity.
 - b. Explicitly construct the underlying data structures: a Hash Table paired with a custom Doubly Linked List.
 - c. Demonstrate that placing an item into a full capacity cache correctly evicts the node at the tail of the list.

21. Why do we use Sentinel (Dummy Head/Tail) nodes in an $O(1)$ LRU cache implementation?
22. What is Belady's Anomaly, and why doesn't standard LRU suffer from it?
23. "How does LFU handle 'Frequency Pollution', and how is it resolved in production systems?
24. Why does Redis use Sampled LRU instead of a true $O(1)$ Hash Map + Doubly Linked List LRU?
25. What is the difference between SLRU (Segmented LRU) and 2Q (Two-Queue) algorithms?
26. Consider the following page access sequence trace:
Sequence Trace = [1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5] **Task:**
 - a. Trace the execution of a **FIFO** cache for capacities $C = 3$ and $C = 4$. Count total cache misses for each and prove that Belady's Anomaly occurs.
 - b. Trace the execution of an **LRU** cache for the exact same sequence across capacities $C = 3$ and $C = 4$. Show that LRU does NOT exhibit Belady's Anomaly.
27. What is the operational difference between max-age and s-maxage in an HTTP Cache-Control header?
28. Why is a Soft Purge safer than a Hard Purge when updating high-traffic assets?
29. How does Edge Compute (e.g., Cloudflare Workers) achieve sub-millisecond cold start times compared to traditional AWS Lambda functions?
30. What is the difference between an ETag and a Last-Modified header during cache revalidation?
31. A global news website hosted in London ($\text{RTT}_{\text{origin}} = 0 \text{ ms}$ for UK users) serves global users in Sydney, Australia. Speed-of-light optical fiber delay between London and Sydney yields a baseline $\text{RTT} = 160 \text{ ms}$. A web page requires 1 base HTML payload + 15 static sub-assets (CSS, JS, images). **Task:**
 - a. Calculate the total network transmission time for a Sydney user downloading the page without a CDN over TCP+TLS 1.3 (assuming 2 RTT setup overhead + sequential asset fetches across 3 connection streams).
 - b. Calculate the page load time when deploying an Edge PoP in Sydney with a 95% static Cache Hit Ratio ($H = 0.95$), where Edge $\text{RTT} = 4 \text{ ms}$.
32. Explain the role of a load balancer in a distributed system. How does it help improve scalability and availability?
33. Explain Weighted Distribution in load balancing with a suitable example. When is it preferred over simple Round Robin?
34. Explain how a load balancer can help prevent overloading of individual application servers.
35. Explain the difference between Layer 4 and Layer 7 load balancing in brief.
36. Explain the advantages of using stateless application servers in a horizontally scaled architecture.

37. Explain the challenges of load balancing stateful services.
38. Compare sticky sessions and stateless services with respect to scalability, fault tolerance, and load balancing.
39. Explain the role and basic working of AWS Elastic Load Balancer (ELB)
40. Explain how a reverse proxy and load balancer can work together in a scalable web application
41. Explain the Cache-Aside (Lazy Loading) strategy with a suitable example.
42. Explain the Write-Through caching strategy and discuss its advantages and limitations.
43. Explain the Write-Back (Write-Behind) caching strategy with a suitable example.
44. Explain the Read-Through caching strategy and compare it with Cache-Aside.
45. Compare Cache-Aside, Write-Through, Write-Back, and Read-Through caching strategies.
46. Explain the concept of cache hit and cache miss with an example.
47. Explain the LRU (Least Recently Used) cache eviction policy with a suitable example.
48. Explain the LFU (Least Frequently Used) cache eviction policy with a suitable example.
49. Explain the FIFO cache eviction policy with a suitable example.
50. Compare LRU, LFU, and FIFO cache eviction policies.
51. Explain why a cache requires an eviction policy when its storage capacity is limited.
52. Explain how a CDN can reduce network latency and improve content delivery performance.